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DATA VISUALIZATION

Interactive Visualization of the Spanish Labor Market

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1 Problem characterization and Application Domain

The first level of design focuses on characterizing the problem and defining the specific challenges the target users may face in decision-making processes.

1.1 Target User and Context

The target users for this tool are **labor market analysts and sociologists** interested in the dynamics of employment and demography in Spain or even the Spanish Public Administration itself (SEPE, regional governments) and economic research institutes.

Possible challenges

These stakeholders operate in a highly complex context where the Spanish labor market exhibits significant structural heterogeneity. The SEPE releases massive datasets containing monthly records of **unemployment, contracts, and job seekers**, broken down by **municipality, gender, age and economic sector**. However, these raw files (CSVs) are fragmented and disconnected. Currently, an analyst cannot easily discern between different behaviours of the economy and sociology. For example whether an increase in unemployment in a specific region is due to a structural crisis or a seasonal effect (e.g., the end of the summer tourist season). Or how each sector has different employment weights depending on the geographical location.

The core motivation behind this tool is the need to contextualize data through a unified control panel that integrates all relevant information, enabling users to discover complex patterns that would otherwise remain hidden without effective visualization.

1.2 Questions and Tasks

Based on the need to tailor policies to regional realities, the tool is designed to answer the following specific questions:

- **Q1: Geographical relevance and temporal evolution of employment in each sector**

"What seasonality patterns in hiring exist across Spain? Do tourism-dependent regions experience sharp peaks in contracts during the summer periods? Where are these regions located? And regarding regions with a heavier reliance on agriculture for employment, do they also follow seasonal patterns? Do they retain workers, or do they hire exclusively for harvest periods?"

These questions are strategically formulated to validate the structural duality and precariousness of the Spanish labor market through the dashboard. The primary goal is to confirm whether the country is sharply divided between a relatively static interior economy and a highly volatile, cyclical periphery.

Specifically, the queries regarding tourism aim to isolate the "Sun & Beach" dependency. The hypothesis here is that coastal regions do not just experience growth, but are dictated by a strict seasonal regime: a massive hiring boom leading up to the summer, followed by a sharp, immediate destruction of employment in the autumn, revealing a vulnerability rather than sustainable strength.

In contrast, the questions focusing on agriculture seek to identify an inverse pattern. It is expected to see provinces where hiring spikes are biologically determined by harvest cycles—such as the winter olive campaign in the south, creating a rhythm that is completely out of sync with the national tourism trends.

Finally, by questioning worker retention, we aim to expose the potentially disposable nature of this labor force. The hypothesis is that these sectors generate extreme turnover with vertical drops in hiring immediately post-season. This would prove that high contract volumes often mask a lack of genuine stability, highlighting an extractive model rather than one based on long-term retention.

- **Q2: Sectoral Vulnerability and Seasonality**

"SOMETHING"

STH

- **Q3: Demographic Pressure (Geospatial Distribution)**

"SOMETHING"

STH

- **Q4: Comparative Regional Benchmarking**

"SOMETHING"

STH

2 Data and Task abstractions

2.1 Dataset Description

This analysis is built over a dataset sourced from the Spanish Public State Employment Service (SEPE). The data is structured into three distinct datasets for each year, covering: [Registered Job Seekers](#), [Registered Unemployment](#) and [Registered Contracts](#).

These datasets offer a high level of granularity, providing a monthly breakdown for every municipality in Spain. Furthermore, the data is stratified by gender, economic sector, and age group, allowing for multidimensional filtering.

The temporal scope of the study spans 15 years, from 2010 to 2025, enabling long-term trend analysis. Additionally, to normalize these absolute values and generate relative metrics (per capita), official population data has been integrated for every Spanish municipality corresponding to the same period.

2.2 Data Justification

To answer the research questions the specific attributes of this dataset are essential for the following reasons:

Q1-Geographical relevance and temporal evolution of employment in each sector

- **Monthly Granularity for Seasonality Detection:** To validate the hypothesis of the "inverse heartbeat" between tourism and agriculture, annual aggregates are insufficient. Monthly data is required to visualize the sharp contrast between the summer peaks (June-July) in tourism-dependent regions ("Sun & Beach") and the winter or autumnal peaks in agricultural provinces (e.g., the olive campaign in January).
- **Sectoral Breakdown:** Distinguishing between the *Services*, *Construction* and *Agriculture* sectors is crucial to isolate the source of employment volatility.
- **The Necessity of Complementary Metrics:** While "Unemployment" measures the stock of jobless individuals, "Registered Contracts" measures the flow and turnover. To answer the question regarding worker retention and precariousness ("Do they retain workers or just hire for harvest?"), we need the volume of contracts.
- **Population Data for Normalization:** To identify geographic patterns, absolute numbers must be normalized. Without population data, large capitals like Madrid would always appear as outliers. Normalizing by population reveals the true intensity of unemployment or hiring in smaller, structurally vulnerable provinces (e.g., the "Empty Spain"), allowing for a fair comparison between regions of vastly different sizes.

3 Interaction and Visual Encoding

To effectively answer the questions and reveal the underlying patterns of the Spanish labor market, the application employs a multi-view strategy.

3.1 Visual Encoding

Q1-Geographical relevance and temporal evolution of employment in each sector

- **Temporal Evolution View (Line & Stacked Areas)**

To reveal the cyclical nature of employment and the seasonal behaviour described in Q1, time is mapped to the X-axis while dataset values are set into the Y-axis. We used **Colors** to differentiate economic sectors (Agriculture, Industry, Services and Construction), a decision justified by the need to immediately discern the specific driver behind a hiring peak. Furthermore, we implemented a **Stacked Area encoding** for the "Relative Distribution" mode. This decision directly addresses the "monoculture" hypothesis: by mapping the Y-axis to percentage share rather than absolute totals, the visualization visually exposes provinces that are structurally dependent on a single sector (e.g., a 90% reliance on Services), which would be less apparent in a standard line chart.

- **Geospatial View (Choropleth Map)**

For the spatial analysis, a Choropleth Map was selected using a sequential **Color Saturation** scale (Yellow-Orange-Red). The critical design decision here was to encode the metric **normalized by population** rather than using absolute values. This is strictly justified by the requirement to identify agricultural hotspots in the sparsely populated "Empty Spain." Without normalization, large metropolitan areas (Madrid, Barcelona) would invariably dominate the visualization due to population density, obscuring the high-intensity seasonal harvest campaigns in smaller provinces that are central to our research questions.

3.2 Interaction Design Decisions

Q1-Geographical relevance and temporal evolution of employment in each sector

- **Temporal Evolution View (Comparative + Filtering)**

Finally, a comparative was integrated to support direct hypothesis validation. By allowing the user to select a primary province and explicitly juxtapose it against a secondary target (e.g., *Jaén vs. Illes Balears*), the design facilitates the direct comparison of divergent patterns. This interaction is essential for confirming the duality of the market, enabling the user to observe on a single axis how one region reaches its peak employment exactly when the other enters its lowest. Also it allows to filter by range of time (to zoom in seasonal patterns).

- **Geospatial View (Dynamic Time Slider)**

To capture the temporal evolution of harvest and tourism seasons, a dynamic time slider with animation was implemented. Static maps are insufficient to represent the temporal

displacement implied by the ‘inverse seasonality’ hypothesis. The animation makes this displacement explicit, allowing the analyst to observe how peaks in hiring shift geographically and providing a dynamic representation of labor market dynamics.

Configuración Global

Indicador:

Contratos Registrados

Filtros Evolución

Provincia Principal:

Madrid

Comparar con (Opcional):

Jaén

Rango de Años:

2010 2021 2024

2010 2012 2014 2016 2018 2020 2022 2024 2025

Escala Temporal:

☐ Anual (Media)

☒ Mensual (Detalle)

☐ Ver distribución % (Áreas Apiladas)

(a) Temporal View Controls

Configuración Global

Indicador:

Contratos Registrados

Filtros Mapa

Escala Temporal:

☒ Anual

☐ Mensual

Año:

2010 2021 2025

2010 2012 2014 2016 2018 2020 2022 2024 2025

Sector a visualizar:

Servicios

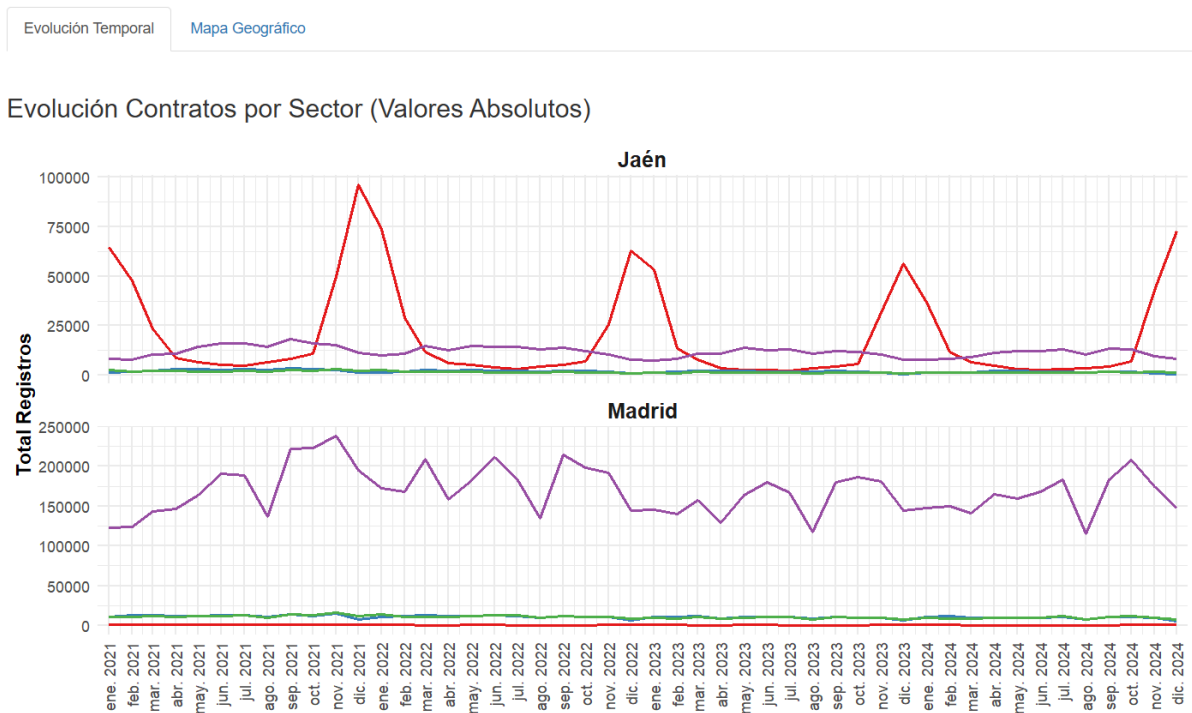
Nota: El valor relativo (%) se calcula sobre la POBLACIÓN TOTAL de la provincia.

(b) Map View Controls

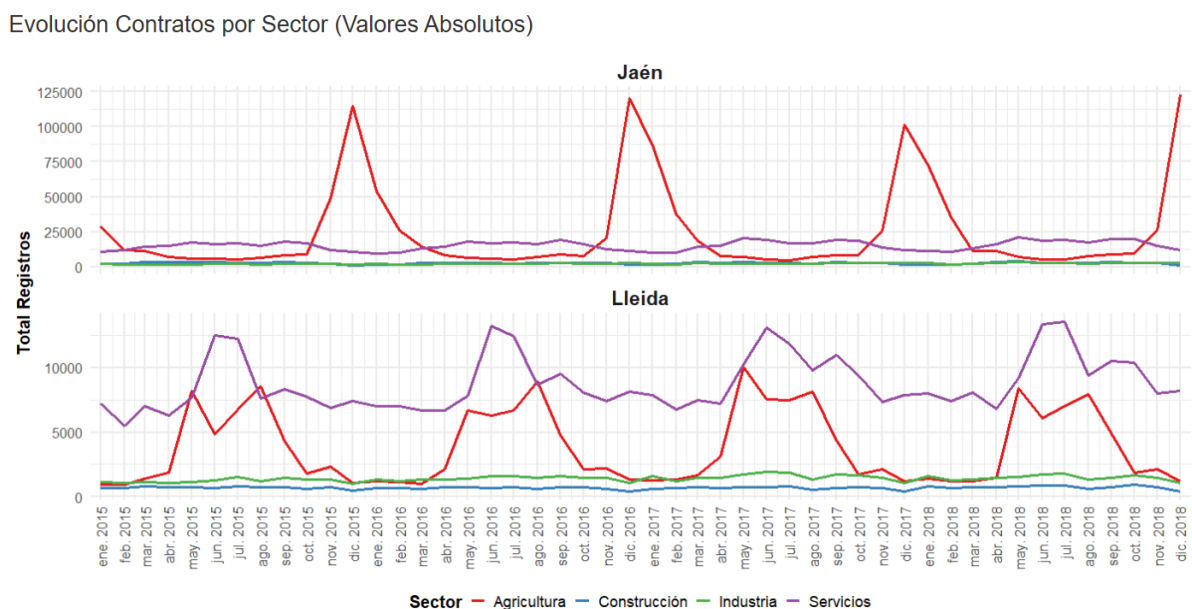
Figure 1: **Interface Design for Interactive Analysis:** Subfigure (a) details the filters for the Temporal View, enabling the user to perform A/B testing between provinces (e.g., Madrid vs. Jaén) and toggle between annual trends and monthly volatility. Subfigure (b) displays the Map View controls, featuring the dynamic time slider (with animation playback) and sector selection, which are essential for tracking the geographical migration of seasonal employment.

3.3 Answers with visual display

Q1-Geographical relevance and temporal evolution of employment in each sector



(a) Volatility vs. Stability: Jaén (Agriculture) vs. Madrid (Services)



(b) Crop Duality: Jaén (Olives) vs. Lleida (Fruit)

Figure 2: **Visualizing Precarity and Retention:** Addressing the critical question *"Do they retain workers?"*, these time-series provide a negative answer for agricultural regions. Subfigure (a) shows Jaén's extreme "sawtooth" pattern (agriculture, red line), where employment verticality implies massive hiring followed by immediate firing, contrasting with Madrid's stability. Subfigure (b) confirms that this volatility is intrinsic to the crop type, with Lleida peaking in summer and Jaen in winter, yet both displaying the same lack of long-term retention.

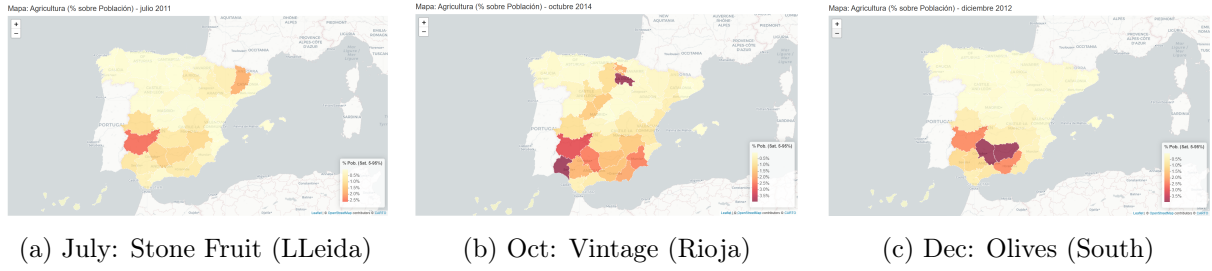


Figure 3: Tracking the Agricultural Seasons: Answering the question regarding agricultural patterns, this sequence validates that agricultural employment is not static. It follows a biological calendar that shifts geographically when each product is collected (intensive in human capital): (a) hiring peaks in the Lleida, Huesca, Extremadura during summer for fruit, (b) moves to the Rioja, Castilla, Murcia in autumn for grapes, and (c) concentrates intensely in the South (Jaen/Cordoba) during winter for olives.

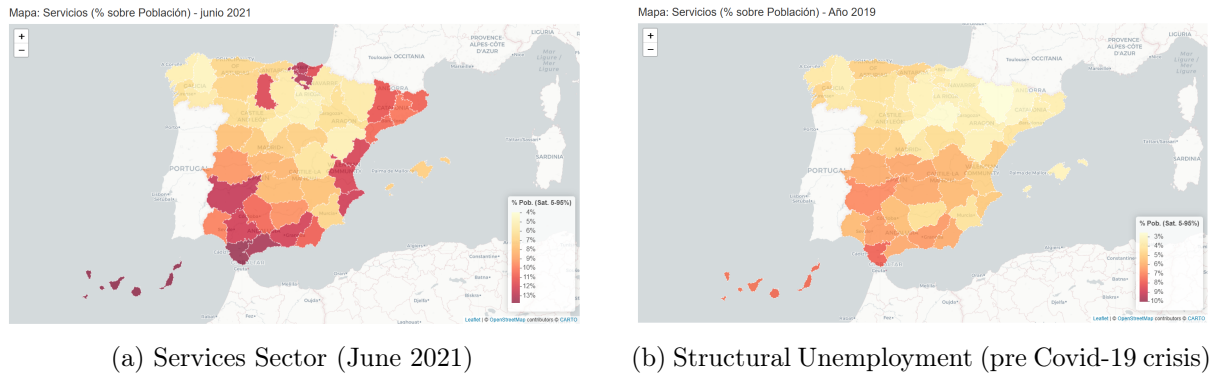


Figure 4: The "Sun & Beach" Effect: Answering the inquiry about tourism-dependent regions, subfigure (a) clearly identifies the coastal belt and islands as high-intensity hiring zones ("hotspots") at the start of summer. This contrasts with subfigure (b), which highlights how this seasonal volatility correlates with long-term structural unemployment in the South.

4 Algorithmic Implementation

The implementation of the visual design relies on specific algorithmic strategies to ensure statistical robustness and interactive fluidity.

Q1-Geographical relevance and temporal evolution of employment in each sector

- **Temporal Evolution Algorithm: Dynamic Normalization**

For the temporal analysis, the system employs a server-side reactive pipeline that executes real-time data transformation based on user input. The critical algorithm handles the switching between "Absolute" and "Relative" modes without requiring pre-processed static files. When the "Relative Distribution" flag is active, the system performs an on-the-fly aggregation: it groups records by date and province, calculates the instantaneous total contract volume, and normalizes each sector's value as a percentage share ($V_{sector} / \sum V_{total}$). This algorithmic approach allows the analyst to instantly assess the "monoculture" hypothesis (structural weight) versus total volume trends within the same view.

- **Geospatial Algorithm: Percentile Clamping**

The mapping implementation addresses the issue of extreme statistical outliers common in labor data (e.g., massive but isolated harvest spikes). A standard linear color scale would result in a "washed out" map where only the outliers are visible. To solve this, we implemented a **Percentile Clamping Algorithm**. The system dynamically calculates the 2nd ($P_{0.02}$) and 98th ($P_{0.98}$) percentiles of the population-normalized distribution. Values falling outside this range are algorithmically "clamped" (saturated) to the minimum or maximum color of the *Yellow-Orange-Red* palette. This ensures that the color gradient remains sensitive to the majority of the data distribution, making subtle seasonal shifts in the "Empty Spain" visible while still acknowledging high-intensity hotspots.

5 Installation and Execution Instructions

To facilitate the reproduction of results and the evaluation of the tool, the application has been designed with a streamlined, self-contained setup pipeline.

5.1 Dependencies

The project relies on standard CRAN packages for data manipulation ('dplyr', 'tidyr'), visualization ('ggplot2', 'scales'), and geospatial rendering ('leaflet', 'sf', 'mapSpain'). Please ensure the following libraries are installed before running the application:

```
install.packages(c("shiny", "dplyr", "ggplot2", "tidyr", "shinyjs",  
                  "scales", "leaflet", "sf", "mapSpain"))
```

5.2 Running the Application

The execution process is simplified to minimize configuration errors. No manual data download is required.

1. Open the project in RStudio.
2. Load the Shiny library in the console:

```
library(shiny)
```

3. Run the application by clicking the **"Run App"** button or executing:

```
runApp()
```

Note on Automated Data Ingestion:

The application includes a self-healing data mechanism. Upon initialization, the system checks the local `/data` directory. If the source files are missing, the app automatically triggers the `preprocessing.R` script to download the raw datasets directly from the SEPE servers, process the population data, and generate the necessary clean dataframes for the 2010–2025 period.